

The Future of HPC in Nuclear Security

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Advances in computational capabilities have transformed national security. For the U.S. nuclear deterrent these advances have allowed wiser options for a production complex that is a fraction of its cold war-era size, enabled more effective military solutions, and have been a safeguard against consequential mistakes. An ability to faithfully simulate increasingly complex physical systems in human learning times has been the key to this progress. Experience spanning almost 80 years shows that qualitative leaps in simulation capabilities have allowed new contributions to national security. Through detailed quantitative analysis of representative simulations, we show that reliance on technologies developed for popular market applications will likely stall progress. We argue that if the United States is to continue benefiting from advances in computing, investments in deeper co-design of hardware and software—addressing levels of branching and sparsity not found in machine learning or most other major market applications—will be needed.

As part of the justification for leasing LANL's first collection of IBM equipment in 1944, J. Robert Oppenheimer estimated that those electromechanical machines would reduce computational times from six or eight months to just three or four weeks and noted: "This saving in time is regarded by me as absolutely essential if the engineering work is to be completed on schedule and is to be designed in accordance with our best knowledge rather than by guess."¹

Since that era, the role of computing in nuclear security has grown as technology has advanced. High-performance computing (HPC) now contributes to virtually all aspects of the U.S. nuclear security mission.

In just the past two decades, the nature of the contributions made by HPC has significantly changed. As late as the 2000s, the U.S. nuclear weapons complex did not rely on computing for assessing implications of subtle features through 3-D calculations. Now it is hard to imagine how the nation could advance the design of inertial confinement fusion ignition capsules or maintain the stockpile without such a capability. These simulations routinely require a half petabyte of

memory, span thousands of compute nodes, and can take many months to complete on the most powerful supercomputers in the world.

Nuclear security applications need to model an extremely large dynamic space and state range in a single coupled application. These applications are characterized by multiple space representations, and coupling disparate spatial and temporal scales. Together, these requirements for simulating the most complex physical systems result in two root challenges: sparsity and branching. These challenges are also important barriers to progress for climate science, astrophysics modeling, seismographic inferences, and aerodynamics design.

Advances in nuclear security applications have been a result of co-design focused on parallel scalability of codes and leveraging Moore's law and Dennard scaling. This has enabled a transition from 1-D single physics models as state-of-the-art in 1980 to 3-D multi-physics models today. As current leakage brought an end to Dennard scaling circa 2003, industry found creative ways to make advances through increased parallelism, greater power density, and specialization. These directions hold the key to continued progress for nuclear security applications. However, because the sparsity and branching demands of these applications are relatively unique, progress will only be made with deep co-design aimed at shaping technologies for the needs of nuclear security.

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CONTRIBUTIONS OF HPC TO NUCLEAR SECURITY

The contributions of computing to nuclear security can be described in the following three broad categories.

Avoiding consequential mistakes: Regarding weapons performance, it is the ability to avoid mistakes that gives the U.S. confidence to maintain a stockpile without testing. Possible mistakes in operations span a broad range: accidents that lead to yield production or plutonium dispersal, pulling back assets from the field for a defect that was actually harmless (or leaving it when we underestimated a defect), miscalculating implications of aging, and so on.

Providing wiser options for the United States: From the perspective of policy, an ability to maintain the stockpile without testing is the most important contribution of computing. From the perspective of time, money, and human effort, computing's most important role may be in allowing the U.S. to take courses of action that it otherwise could not. To underscore this point, it is worth noting a view that was held by thoughtful experts early in the stewardship program:

"If they are remanufactured to the same specifications ...they will be as good as the day they were first made. This can be done any number of times, and is the basis for my confidence in the future stockpile."²

We now know that a program of strict remanufacture is infeasible, and that computational capabilities allowing the nation to innovate in manufacturing are required to maintain the deterrent. Computing has come to function as the kind of guardrail that robust production and nuclear testing once provided.

Enabling more effective military solutions: This relates to the ability to design and field weapons that hold particular targets at risk, to mitigate threats to weapons, and the time required for fielding responses to emerging developments.

There are many examples highlighting the value of computing advances in these areas. Enabling the use of cast pits instead of reproducing the rocky flats wrought process relied on large suites of uncertainty quantification calculations. Developing the predictive capability to move away from an empirical approach for designing detonators depends on higher fidelity and 3-D multiphysics calculations. Other examples include confidence in maintaining aged pits and confidently accepting certain defects found during weapon system surveillance. These capabilities have resulted in hundreds of millions of dollars in savings while cutting years off of schedules.

An important dimension of advances in computing relates to human timescales for learning and

productivity. These timescales set basic limits on the type of contribution that can be made with a given level of technology. As a rule of thumb, simulations that are conducted by a fair fraction of staff as part of their regular work must require less than a week or so to complete, though one business day is ideal. Longer running times interfere with an individual's productivity and ability to learn through iteration. In a handful of cases, a problem has been so important that it warranted an extraordinary commitment of time. Empirically, evidence over the last 15 years suggests that the upper limit appears to be devoting about half of a world-class machine to a problem for six months, something we now do with regularity. With later generations of technology, these heroic calculations can become part of the normal workflow. We have seen these order of magnitude improvements in time to solution in prior times of technology shift, such as the serial to vector era, the rise of the microcomputers, transitions to distributed memory clusters, and most recently the prevalence of GPU computing. The future of tailored and specialized computing must now be embraced.

VALUE OF FUTURE ADVANCES

Given the lessons of history, how should we reason about the value of future advances in computing? One approach focuses on raw computing power, such as floating point operations per second (FLOPS). From the perspective of being able to do more for security, the advances associated with an ability to tackle greater complexity, or a given complexity in a human timescale, are much more important. Transformational improvements in this regard have tended to fall into just a few categories. We illustrate these in the context of national ignition facility (NIF) capsule simulations, owing to their wide discussion in the academic literature.

Multiscale: An ability to model systems where disparate length- and time-scales are important. With recent advances, it has been possible to show that the tiny, few-micron-wide fill tube used for putting Deuterium-Tritium into the center of the capsule can disrupt the fuel and degrade performance.³ Computing demands do not scale well with spatial resolution; a 10-fold increase in resolution requires 10,000 times more computing for 3-D problems. Recently, a single, reasonably faithful 3-D model of a capsule in the MARBLE campaign, as illustrated in Figure 1, is already pushing the limits of what is possible with a run time of nearly a year and a half on a quarter of the Sierra supercomputer, which has over 4000 compute nodes each with four NVIDIA Tesla GPUs and two IBM

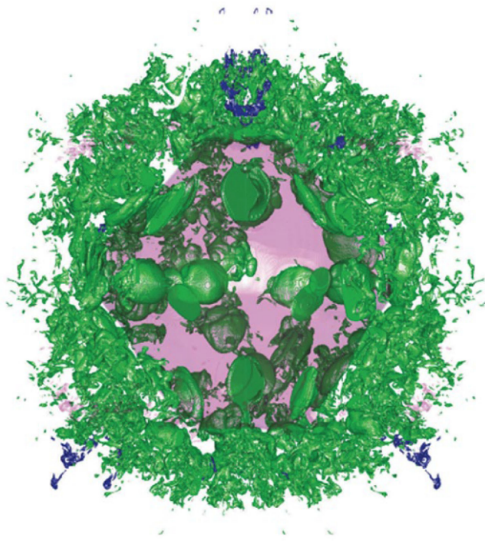


FIGURE 1. Snapshot of a simulation of an NIF shot in the MARBLE campaign showing the shock (magenta) interacting with pores (green) and leading to material mixing.⁴ This figure is reproduced with permission from Albright et al., Copyright 2022 Authors, licensed under a Creative Commons Attribution (CC BY) license.

Power9 CPUs. A 3-D simulation that resolves finer-scale engineering features is well beyond current capabilities, but would provide invaluable insight into the behavior of these complex systems.

Phase-space explosion: Adopting higher fidelity models often means taking on more complexity. In NIF hohlraums, rapid laser energy deposition drives the system out of local thermodynamic equilibrium, opening up a new dimension of phase space to represent the atomic level populations. We can now model this effect approximately but the weakly collisional plasma blown off by the hohlraum now demands a model with twice the number of phase space dimensions, exponentially increasing complexity. Accurate modeling of this is not yet feasible even in 2-D.

These categories provide a useful framework for reasoning about the significance of future advances in computing. Of potential transformative value are the ability to:

- › complete, in a week, complex 3-D simulations that presently take most of a year on our most capable computers allowing us to address uncertainties and to underwrite important decisions for the stockpile;
- › link details of the manufacture of materials and parts to performance allowing us to test the hypothesis that fine details of the capsule

microstructure are key to performance in ignition capsule implosions;

- › accurately simulate the chain of events going from nuclear device performance, to outputs in an environment, to effects on complex structures and components;
- › predictively model fine-scale instabilities and their influence on the behavior of a macroscopic system. This has important implications for nuclear security, climate and weather modeling, and fusion energy.

The ability to realize these transformative capabilities in the era of hardware tailoring and specialization will require a deeper level of co-design that focuses our efforts on the root challenges we face in our applications. The following section describes these root challenges and how they differ from those exhibited by other applications that are currently driving major advances in HPC.

TWO ROOT CHALLENGES IN APPLICATIONS

Nuclear security applications are distinguished by the complexity of coupled multiphysics, high resolution, many materials, and multiple space representations within a single application. The state-of-the-art simulations require 1/2 petabyte of memory and the use of sparse data structures, such as adaptive mesh refinement (AMR). Algorithms adapted to these data structures result in sparse memory access. Over 50% of all instructions in these applications are sparse memory operations resulting from indirection to access AMR leaf cells. Roofline analysis further shows that all routines are either mostly or completely main-memory bound (0.001–0.2 FLOP/byte).

Branching in nuclear security applications is the result of a number of factors. Sparse data structures result in fine-grained branching in kernels using these structures. Multiple levels of indirection results in data hazards that must be guarded by branching or atomics. Finally, Monte Carlo (MC) algorithms have fine-grained branching of different interactions of a particle through a domain. This branching can be seen in the relatively high (25) percentage of conditional instructions in these applications.

Sparsity and branching are not well represented by other HPC workloads. Graph analytics are at the far end of sparsity. Deep neural networks (DNNs) exhibit dense memory operations and even those workloads with some sparsity are still orders of magnitude denser than ours.⁵ DNN workloads have little branching.⁶ Some

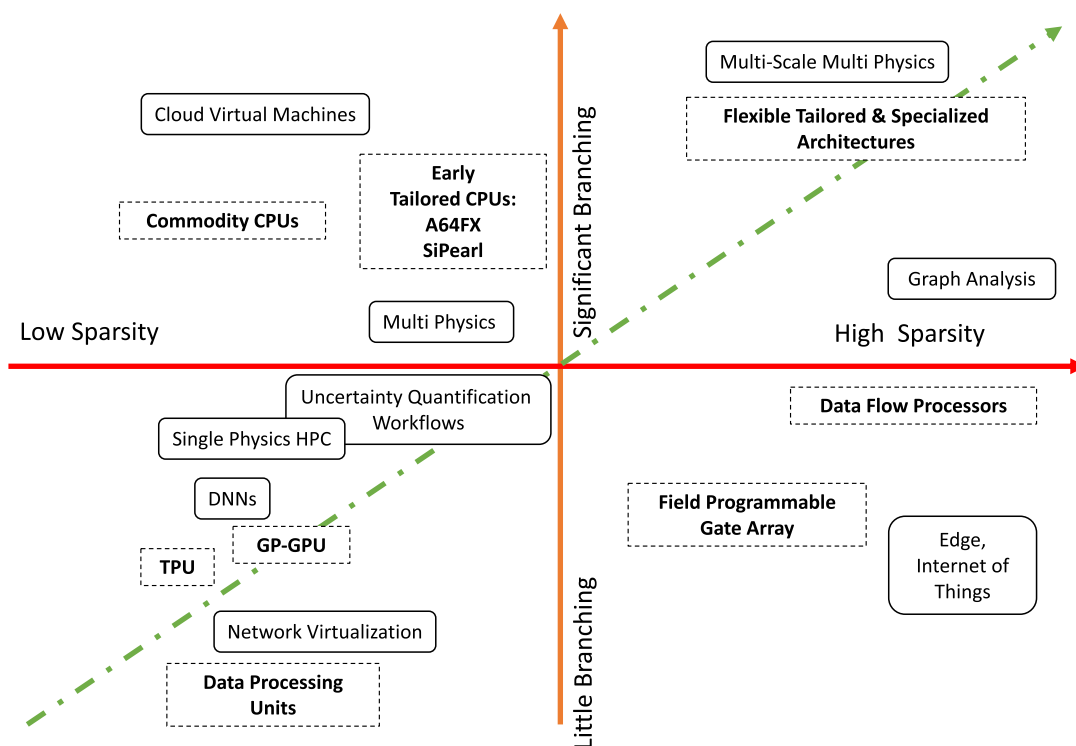


FIGURE 2. Two root challenges of sparsity and branching. Technologies illustrated in dashed rectangles and workloads in solid rounded corner rectangles. Commodity CPU technologies are better suited to more branching in code but still struggle with higher sparsity. GPUs are optimized for low sparsity and little branching.

workloads are similar in sparsity to nuclear security applications, such as database join operations and there are initial efforts focused on accelerating them through deep co-design.⁷

TOWARD A TAXONOMY OF POSSIBLE SOLUTIONS

When considering solutions to the two root challenges it is important to consider the broad technology marketplace and the characteristics of the computing problems that drive this market. Figure 2 provides a simple way to characterize computing problems and the technologies used to solve them. One axis describes branching where higher levels of branching correspond to cases in which future operations depend on earlier ones. General purpose CPUs are designed for this with deep out of order pipelines and speculative execution. GPUs and TPUs are not and can suffer poor performance. The second axis describes sparsity of memory accesses where problems with low sparsity are characterized by large contiguous memory accesses that achieve high bandwidth. As sparsity increases, memory accesses are no longer as contiguous and result in low bandwidth.

DNN applications are characterized by low sparsity/little branching and perform exceedingly well on GPUs or TPUs. As branching increases, CPUs become a more appropriate technology. When we consider sparsity, the solution space is less clear. Workloads with high sparsity and moderate branching can benefit from data flow processors. Complex workloads using sparse data structures with branching expose a gap in technologies (upper right).

Given this landscape, there are four paths to address the needs of complex workloads.

Rely on Market-Driven Advances in CPUs

This initially appears to be an attractive option but our highest priority is to reduce time to solution for simulations that presently take months using a sizable fraction of world-class machines. The considerations of some major milestones in the history of nuclear weapons simulation shows why these needs will not be met with reliance on market-driven advances in CPUs. The Dennard scaling era (1986 through 2003) was highly productive with advances in resolution and physics fidelity. In 1995, the first partial-domain low-resolution 3-D simulation for

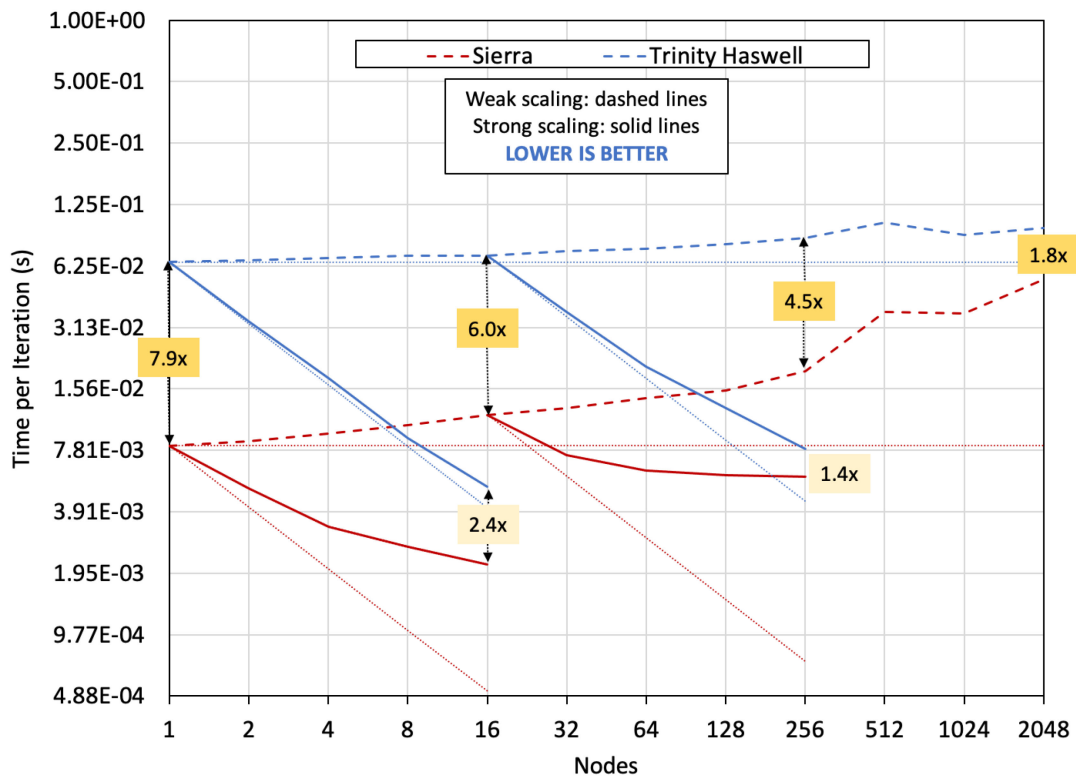


FIGURE 3. Weak and strong scaling of the PartiSN discrete ordinates transport application. Image courtesy R. Joseph Zerr (LANL).

understanding a particular system of concern became possible. In the decade following that, peak clock rates increased by a factor of 10, DRAM densities 40-fold, and the performance of CPUs increased by 50% each year. Thanks to those advances it took only another five years for Los Alamos scientists to conduct the first 3-D large-domain (low resolution) simulations of a comparable system. Then progress began to stall. It was more than a decade later that the first higher fidelity partial-domain (2012), and moderate resolution large-domain (2017), simulations could be completed.

Move Our Applications With the Market

Another approach is to move our codes to technologies designed for other markets. The most viable being the GPU. In many domains, this approach has been successful. Density functional theory⁸ with dense computations is one example. HPC codes which lack computational density can still see improvements due to high bandwidth memory (HBM) on the GPU. As observed on the Sierra platform at LLNL: "Performance is heavily tied to the HBM associated with the GPUs. The more data we can fit into that type of memory, the better our codes will perform."⁹ The GPU architecture and associated

programming model can exploit the wide parallel interface of HBM when sufficient work (data) is available. As the amount of work (concurrent memory operations) falls off, performance improvements over CPUs diminish.¹⁰ Other applications can exhibit faster fall off in strong scaling such as the LAMMPS molecular dynamics LJ Benchmark where CPU and GPU performance are equal at 64 nodes. Predictive wind plant modeling has experienced¹¹ even weak scaling fall off more quickly on GPUs. Strong scaling past tens or a few hundred nodes can be challenging on GPUs, particularly for applications with internode synchronization, such as wavefront transport. The much higher parallelism of the GPU pushes codes into the strong scaling limit where Amdahl's law dominates. Figure 3 illustrates strong scaling on GPUs falling off at small node counts. Even weak scaling can be problematic; on a single Quad Nvidia Volta GPU node the performance of PartiSN¹² is 7.9x higher than on a single Dual Intel Haswell CPU node, but at 2048 the difference is only 1.8x.

Develop Entirely New Computational Methods

It is expensive and time consuming to make an application with particular characteristics work on a technology

designed for different characteristics. The traditional approach to MC transport has a large amount of branching. If you try to use the standard algorithms on GPUs, the performance is abysmal. To get good performance, an enormous effort has gone into implementing clever new algorithms with reduced sparsity and branching. Though the first papers on this topic appeared more than a decade ago, major code projects are still not done with implementation. Others have chosen to implement new codes given the algorithmic differences.¹³

Another approach might be moving entire application domains to radically different methods, such as ML. One example is the use of DNNs to accelerate climate science data analysis on Summit at the OLCF¹⁴ resulting in a 2018 Gordon Bell. While encouraging, the use of ML to replace core hydrodynamics methods or radiation transport methods is nascent and there are currently no credible alternatives to traditional approaches. One promising area is the use of machine learning to train models on high-fidelity physics, such as combustion chemistry or grain structure aware strength modeling.

Agile Deep Co-design

Co-design is based on the interplay between progress in computational methods and evolving computer architectures. To date, we have mostly been engaged in what has been characterized as “hardware-driven algorithm design.”¹⁵ The result has been a fair amount of success in modifying or adopting new methods to make use of technologies developed for quite different purposes, and that are in some ways inimical to the computational patterns in our workloads.

If we consider an alternative, where the design of the processor, memory subsystem, and accelerators can be tailored or specialized to needs of multiphysics, solutions to our two root challenges may be attainable. Due to a confluence of factors, there are grounds for optimism that we can move to a model of co-design in which hardware choices are shaped to better meet our needs.¹⁶ In fact, an example of a tailored CPU for HPC has been built, as represented by the Fujitsu A64FX. These processors have incorporated HBM-2 memories providing an 8x bandwidth improvement over comparable CPUs of the same era. The introduction of the scalable vector extension in A64FX provides high FLOP performance with support for variable vector widths and gather/scatter (G/S) operations. The European processor initiative has embarked on a similar path of deeper co-design.¹⁷

A major hurdle to overcome in this approach is the lack of agility in deeper co-design. This is the result of

many factors, part is the historic mismatch in iteration timescales at different levels of the co-design “stack.” At the algorithmic level, a computational scientist may iterate very quickly with a computer scientist or applied mathematician to test different approaches in a sparse linear solver for instance. Iterations may be measured in days. If one includes changes to an instruction set to improve this algorithm, the iteration time can balloon into months or even years. This is a result of many factors, part of which is attributable to the complexity involved in making or simulating such a hardware change and the resulting changes required in compilers and software before an application can even begin experimentation. Other challenges are cultural, where computer scientists, engineers, computational scientists, and physicists each have their own language and culture that can impede communication.

To overcome the latter challenge, the Department of Energy (DOE) established major initiatives; the Accelerated Strategic Computing Initiative was established in 1998 and the Advanced Scientific Computing Research program established computational co-design centers in 2010. These and many other similar projects and initiatives have improved the community’s ability to communicate. Deeper engagement with hardware architects will require bridging yet another vernacular, but the community is positioned to do so.

The former challenge is more acute, there is a significant need for tools and techniques to shape applications and hardware in tandem. Recent work has focused on identifying common bottlenecks across nuclear security codes that are amenable to hardware tailoring. One promising target is the memory subsystem. Figure 4 shows a bottleneck analysis across a variety of codes including FLAG (hydrodynamics), PartiSN (discrete ordinates transport), JAYENNE (implicit MC transport), and xRAGE (hydrodynamics). This analysis alone is only partially actionable, perhaps indicating that improved bandwidth would prove beneficial. Deeper analysis is required to identify more granular architectural features and how we might redesign aspects of our codes to best utilize them.

We know that the data structures and methods in codes lead to sparse memory access and we can capture memory access patterns (both temporal and spatial) using program instrumentation and data analysis. These patterns can be used for co-design of advanced memory technologies, such as multilevel G/S and 2-D or 3-D addressing schemes alongside algorithmic changes. The use of block-structured AMR rather than the current cell-based AMR in xRAGE could significantly reduce sparsity and improve performance. To enable this exploration, we have developed more agile prototyping

	Memory subsystem						Floating Point		
	L1	L2	L3	DRAM	DRAM BW	Mem Latency	DP FLOPs	Vectorization	Non-FP
Flag 3D ALE AMR							3.70%	11.20%	96.30%
Flag 3D ALE Static							2.20%	7.20%	97.80%
xRAGE 3D AMR							6.50%	14.00%	93.50%
PartiSN 42 Groups							26.20%	90.40%	72.80%
Jayenne DDMC Holraum							15.50%	0.00%	84.50%

FIGURE 4. Heatmap showing application bottlenecks. Red indicates a significant bottleneck with gradation to green indicating no bottleneck to application performance. Blue highlights the percentage of non-FLOP operations.

environments for alternative data structures. This has enabled explorations in months rather than years.

The potential for deeper co-design with HPC applications has been realized in the past. The Cray X1 supported G/S to a large register file; for sparse STREAM Triad calculations, performance was an order of magnitude higher compared to the IBM Power4. More recent work has begun to revisit memory subsystem design for G/S.^{18,19} These efforts are important and may ultimately influence hardware design. An excellent example of more agile deep co-design is work²⁰ that undertook analysis of workloads in graph analytics, databases, and ML to identify common computational patterns. This led to the design of a sparse processing unit in Chisel, simulated in Gem5, and taped out in a 28-nm process node. Algorithms were then redesigned for this architecture. The results show major performance improvements over CPU and GPU architectures. This rapid co-design cycle was not even possible just a few years ago until the commoditization of processor design and fabrication.

CONCLUSION

In this article, we have presented the two root challenges of sparsity and branching exhibited by nuclear security applications. Broader market forces are pushing technologies away from a solution space to these challenges. Addressing this divergence could take a number of paths. Relying upon advances in general purpose processing alone will result in continued diminishing improvements in simulation capabilities, effectively freezing our capabilities in this decade. Moving our applications to technologies that are driven by the broader market has thus far shown only marginal improvements at the scale at which our most complex applications must run. Developing entirely new methods that better map to technology driven by the broader market can take a decade or longer to make their way into production multiphysics codes and it is

unclear if new methods will yield significant improvements at scale. Future advances in computational capabilities for nuclear security and the U.S. will therefore require a deeper level of co-design that focuses on tailoring and specialization of computing hardware alongside algorithmic advances. Forward progress in this regard will be governed by our ability to do this in an agile and coordinated way spanning national laboratories, academia, and industry.

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REFERENCES

1. H. Bethe et al., "IBM calculations of implosion hydrodynamics," Tech. Rep. Los Alamos National Laboratory, 1944.
2. "Final review of the comprehensive nuclear test ban treaty (treaty doc 105-28)," vol. 4.
3. B. M. Haines et al., "Robustness to hydrodynamic instabilities in indirectly driven layered capsule implosions," *Phys. Plasmas*, vol. 26, 2019, Art. no. 012707, doi: [10.1063/1.5080262](https://doi.org/10.1063/1.5080262).
4. B. J. Albright et al., "Experimental quantification of the impact of heterogeneous mix on thermonuclear burn," *Phys. Plasmas*, vol. 29, 2022, Art. no. 022702.
5. T. Gale et al., "Sparse GPU kernels for deep learning," in *Proc. Int. Conf. High Perform. Comput., Netw., Storage Anal.*, 2020, pp. 1–14.
6. Y. Yu et al., "Dynamic control flow in large-scale machine learning," in *Proc. 13th EuroSyst. Conf.*, 2018, vol. 18, pp. 1–15.
7. O. Polychroniou et al., "Rethinking SIMD vectorization for in-memory databases," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2015, pp. 1493–1508.

8. A. Ziogas et al., "A data-centric approach to extreme-scale ab initio dissipative quantum transport simulations," in *Proc. Int. Conf. High Perform. Comput., Netw., Storage Anal.*, 2019, pp. 1–13.
9. C. Clouse, "The sierra era." [Online]. Available: <https://wci.llnl.gov/sierra-era>
10. J. Vincent et al., "Strong scaling of OpenACC enabled Nek5000 on several GPU based HPC systems," in *Proc. Int. Conf. High Perform. Comput. Asia-Pacific Region*, 2022, pp. 94–102.
11. M. Sprague et al., "ExaWind: Exascale predictive wind plant flow physics modeling," Sandia National Lab. (SNL-NM), Albuquerque, NM, USA; Sandia National Laboratories, Livermore, CA, USA, Tech. Rep. SAND2019-0178C, 2019.
12. R. E. Alcouffe and R. S. Baker, "Time-dependent deterministic transport on parallel architectures using PARTISN," Los Alamos National Lab., Los Alamos, NM, USA, Tech. Rep. LA-UR-98-272; CONF-980403, 1998.
13. M. Balazs et al., "A GPU-based direct monte carlo simulation of time dependence in nuclear reactors," *Ann. Nucl. Energy*, vol. 132, pp. 46–63, 2019.
14. T. Kurth et al., "Exascale deep learning for climate analytics," in *Proc. Int. Conf. High Perform. Comput., Netw., Storage Anal.*, 2018, pp. 649–660.
15. J. Shalf, "The future of computing beyond Moore's law," *Philos. Trans. Roy. Soc. A*, vol. 378, no. 2166, 2020, Art. no. 20190061.
16. J. Dean, D. Patterson, and C. Young, "A new golden age in computer architecture: Empowering the machine-learning revolution," *IEEE Micro*, vol. 38, no. 2, pp. 21–29, Mar./Apr. 2018.
17. M. Kovač, "European processor initiative: The industrial cornerstone of EuroHPC for exascale era," in *Proc. 16th ACM Int. Conf. Comput. Front.*, 2019, Art. no. 319.
18. V. Seshadri et al., "Gather-scatter DRAM: In-DRAM address translation to improve the spatial locality of non-unit strided accesses," in *Proc. 48th Int. Symp. Microarchitecture*, 2015, pp. 267–280.
19. N. Tanabe et al., "A memory accelerator with gather functions for bandwidth-bound irregular applications," *Proc. 1st Workshop Irregular Appl.: Architectures Algorithms*, 2011, pp. 35–42.
20. V. Dadu, J. Weng, S. Liu, and T. Nowatzki, "Towards general-purpose acceleration: Finding structure in irregularity," *IEEE Micro*, vol. 40, no. 3, pp. 37–46, May/Jun. 2020.

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